Mixed effects analysis for seizure trigger data, accounting for half-lives of ASMs

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This script uses a moving average of 90 days in combination with missed doses of ASMs on the last day of the window to look at the relationship with seizures on day window+1 (i.e., day 91 for the first window). Notably, this script takes into account the ASM half life.

Format data for excel spreadsheet

Load file

```
filename = 'all_patient_diaries.xlsx';

% Extract information about number of sheets in the excel
[~, sheetNames] = xlsfinfo(filename);

% Load the data from each sheet/participant as a separate variable
for i = 1:length(sheetNames)
    sheetData = xlsread(filename, sheetNames{i});
    varName = matlab.lang.makeValidName(sheetNames{i});
    assignin('base', varName, sheetData);
end

% Extract only the seizure data (i.e., column 2) and place it all into one
% variable called seizure where each column is a different participant
for i = 1:length(sheetNames)
    sheetData = xlsread(filename, sheetNames{i});
    seizure(:,i)=sheetData(:,2);
end

% As the raw data is in half days, let's shrink the temporal data such that
% it goes by days rather than half days. The below commands just takes into
% account whether there was a seizure/no seizure for a given day. For
% example, if someone had 3 seizures in a given day, under this format they
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```
% would just be counted as having 1 seizure day for that day.
[numRows, numCols] = size(seizure);

seizure_days = zeros(numRows / 2, numCols);
    for row = 1:numRows/2
        for col = 1:numCols
            % Check if there is a positive number in the two consecutive rows
            if seizure(2*row-1, col) > 0 || seizure(2*row, col) > 0
                seizure_days(row, col) = 1;
            end
        end
    end
```

Calculate MA for a 90 day window

Essentially within a 90 day sliding window, this takes the number of seizure days divided by 90 for an estimate of seizure risk.

```
for i = 1:191
    for j = 1:27
        MA(i,j)=sum(seizure_days(i:i+89,j));
        MA_90(i,j)=MA(i,j)/90;
    end
end

% Since we are looking at the relationship between MA and seizures on day
% sliding window + 1, the first day would be day 91 since the MA requires
% 90 days of data. Let's create a matrix of seizure days that only starts
% at day 91 for further analysis later on.
seizure_days_forecast=seizure_days(91:281,:);
```

Extract the ASM data for each subject

End of data format will be 3D matrix [Dose X ASM X Subject]. Essentially this section will mark the timepoints (with a 1) when the number of consecutive doses missed exceeds one half life for the ASM missed. This will then reset if the medication is re-dosed.

```
ASM_HL=zeros(562,6,27);

% Patient 4165: Column 3 Keppra, Column 4 Epidiolex, Column 5 Fluramine
ASM(:,1:3,1)=BIDMC4165(:,3:5);
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,1))
    % Check if the current number is negative
    if ASM(i,1,1) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
```

```
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,1) = 1;
    end
end
% Epidiolex has a max half life of 61 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,1))
    % Check if the current number is negative
    if ASM(i,2,1) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,1) = 1;
    end
end
% Flenfluramine has a max half life of 30 hours, so if there are 2 missed
doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 2;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,1))
    % Check if the current number is negative
    if ASM(i,3,1) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,1) = 1;
    end
end

% Patient 4216: Column 3 CBD oil, Column 4 Lamotrigine
ASM(:,1:2,2)=BIDMC4216(:,3:4);
% Epidiolex has a max half life of 61 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,2))
    % Check if the current number is negative
```

```
    if ASM(i,1,2) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,2) = 1;
    end
end
% Lamotrigine has a max half life of ~33 hours so if there are 2 missed
% doses consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 2;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,2))
    % Check if the current number is negative
    if ASM(i,2,2) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,2) = 1;
    end
end

% Patient 4251: Column 3 CBD oil, Column 5 Onfi
ASM(:,1,3)=BIDMC4251(:,3);
ASM(:,2,3)=BIDMC4251(:,5);
% Epidiolex has a max half life of 61 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,3))
    % Check if the current number is negative
    if ASM(i,1,3) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,3) = 1;
    end
end
% Onfi has a max half life of 42 hours, so if there are 3 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
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```
consecutive_limit = 3;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,3))
    % Check if the current number is negative
    if ASM(i,2,3) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,3) = 1;
    end
end

% Patient 4417: Column 5 Keppra, Column 6 Lamictal XR, Column 7 Vimpat
ASM(:,1:3,4)=BIDMC4417(:,5:7);
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;               % To keep track of consecutive negative numbers
consecutive_limit = 1;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,4))
    % Check if the current number is negative
    if ASM(i,1,4) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,4) = 1;
    end
end
% Lamictal XR has a max half life of ~33 hours, so if there are 2 missed
doses
% consecutively, then at risk of being at ~50% concentration.
% Loop through each element of the column
% Initialize variables
count = 0;               % To keep track of consecutive negative numbers
consecutive_limit = 2;   % Set the limit for consecutive negative numbers
for i = 1:length(ASM(:,2,4))
    % Check if the current number is negative
    if ASM(i,2,4) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,4) = 1;
    end
end
```

```
% Vimpat has a max half life of ~13 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,4))
    % Check if the current number is negative
    if ASM(i,3,4) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,4) = 1;
    end
end

% Patient 4614: Column 3 Aptiom, Column 4 Xcopri
ASM(:,1:2,5)=BIDMC4614(:,3:4);
% Aptiom has a max half life of ~20 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,5))
    % Check if the current number is negative
    if ASM(i,1,5) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,5) = 1;
    end
end
% XCopri has a max half life of ~60 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 5;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,5))
    % Check if the current number is negative
    if ASM(i,2,5) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
```

```
        ASM_HL(i,2,5) = 1;
    end
end


% Patient 4684: Column 5 Phenobarb, Column 6 Lamotrigine, Column 8
% Levetiracetam
ASM(:,1:2,6)=BIDMC4684(:,5:6);
ASM(:,3,6)=BIDMC4684(:,8);
% Phenobarb has a max half life of ~118 hours, so if there are 9 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 9;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,6))
    % Check if the current number is negative
    if ASM(i,1,6) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,6) = 1;
    end
end
% Lamictal has a max half life of ~33 hours, so if there are 2 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Loop through each element of the column
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 2;     % Set the limit for consecutive negative numbers
for i = 1:length(ASM(:,2,6))
    % Check if the current number is negative
    if ASM(i,2,6) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,6) = 1;
    end
end
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,6))
    % Check if the current number is negative
    if ASM(i,3,6) < 0
```

```
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,6) = 1;
    end
end

% Patient 4711: Column 3 Felbatol, Column 4 Zonisamide, Column 5 Lamictal
ASM(:,1:3,7)=BIDMC4711(:,3:5);
% Felbatol has a max half life of 23 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,7))
    % Check if the current number is negative
    if ASM(i,1,7) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,7) = 1;
    end
end
% Zonisamide has a max half life of 63 hours, so if there is 5 missed doses
then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,7))
    % Check if the current number is negative
    if ASM(i,2,7) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,7) = 1;
    end
end
% Lamictal has a max half life of ~33 hours, so if there are 2 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Loop through each element of the column
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
```

```
consecutive_limit = 2;   % Set the limit for consecutive negative numbers
for i = 1:length(ASM(:,3,7))
    % Check if the current number is negative
    if ASM(i,3,7) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,7) = 1;
    end
end

% Patient 4722: Column 3 Felbatol, Column 4 Onfi, Column 6 Depakote, Column
% 7 Gabapentin, Column 8 Zonisamide, Column 9 Fenfluramine
ASM(:,1:2,8)=BIDMC4722(:,3:4);
ASM(:,3:6,8)=BIDMC4722(:,6:9);
% Felbatol has a max half life of 23 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;               % To keep track of consecutive negative numbers
consecutive_limit = 1;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,8))
    % Check if the current number is negative
    if ASM(i,1,8) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,8) = 1;
    end
end
% Onfi has a max half life of 42 hours, so if there are 3 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;               % To keep track of consecutive negative numbers
consecutive_limit = 3;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,8))
    % Check if the current number is negative
    if ASM(i,2,8) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,8) = 1;
    end
end
```

```
% Depakote has a max half life of 16 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,8))
    % Check if the current number is negative
    if ASM(i,3,8) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,8) = 1;
    end
end
% Gabapentin has a max half life of 7 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,4,8))
    % Check if the current number is negative
    if ASM(i,4,8) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,4,8) = 1;
    end
end
% Zonisamide has a max half life of 63-69 hours, so if there is 5 missed
doses then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,5,8))
    % Check if the current number is negative
    if ASM(i,5,8) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,5,8) = 1;
    end
```

```
end
% Flenfluramine has a max half life of 30 hours, so if there are 2 missed
doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 2;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,6,8))
    % Check if the current number is negative
    if ASM(i,6,8) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,6,8) = 1;
    end
end

% Patient 4768: Column 3 Topomax, Column 4 Epidiolex, Column 5 Lamictal,
% Column 6 Zarontin
ASM(:,1:4,9)=BIDMC4768(:,3:6);
% Topomax has a max half life of ~21 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,9))
    % Check if the current number is negative
    if ASM(i,1,9) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,9) = 1;
    end
end
% Epidiolex has a max half life of 61 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 5;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,9))
    % Check if the current number is negative
    if ASM(i,2,9) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
```

```
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,9) = 1;
    end
end
% Lamictal has a max half life of ~33 hours, so if there are 2 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Loop through each element of the column
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 2;     % Set the limit for consecutive negative numbers
for i = 1:length(ASM(:,3,9))
    % Check if the current number is negative
    if ASM(i,3,9) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,9) = 1;
    end
end
% Zarontin has a max half life of ~60 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,4,9))
    % Check if the current number is negative
    if ASM(i,4,9) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,4,9) = 1;
    end
end

% Patient 4835: Column 3 Oxtellar, Column 4 Epidiolex, Column 5 Clobazam,
Column 9 Fintepla
ASM(:,1:3,10)=BIDMC4835(:,3:5);
ASM(:,4,10)=BIDMC4835(:,9);
% Oxtellar has a max half life of ~11 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,10))
```

```
    % Check if the current number is negative
    if ASM(i,1,10) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,10) = 1;
    end
end
% Epidiolex has a max half life of 61 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,10))
    % Check if the current number is negative
    if ASM(i,2,10) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,10) = 1;
    end
end
% Onfi has a max half life of 42 hours, so if there are 3 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 3;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,10))
    % Check if the current number is negative
    if ASM(i,3,10) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,10) = 1;
    end
end
% Flenfluramine has a max half life of 30 hours, so if there are 2 missed
doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 2;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
```

```
for i = 1:length(ASM(:,4,10))
    % Check if the current number is negative
    if ASM(i,4,10) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,4,10) = 1;
    end
end

% Patient 4982: Column 3 Lamotrigine, Column 4 Banzel, Column 5 Epidiolex
ASM(:,1:3,11)=BIDMC4982(:,3:5);
% Lamictal has a max half life of ~33 hours, so if there are 2 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Loop through each element of the column
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 2;    % Set the limit for consecutive negative numbers
for i = 1:length(ASM(:,1,11))
    % Check if the current number is negative
    if ASM(i,1,11) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,11) = 1;
    end
end
% Rufinamide has a max half life of ~10 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Loop through each element of the column
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
for i = 1:length(ASM(:,2,11))
    % Check if the current number is negative
    if ASM(i,2,11) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,11) = 1;
    end
end
% Epidiolex has a max half life of 61 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
```

```
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 5;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,11))
    % Check if the current number is negative
    if ASM(i,3,11) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,11) = 1;
    end
end

% Patient 5057: Column 4 Keppra, Column 6 Onfi
ASM(:,1,12)=BIDMC5057(:,4);
ASM(:,2,12)=BIDMC5057(:,6);
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 1;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,12))
    % Check if the current number is negative
    if ASM(i,1,12) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,12) = 1;
    end
end
% Onfi has a max half life of 42 hours, so if there are 3 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 3;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,12))
    % Check if the current number is negative
    if ASM(i,2,12) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,12) = 1;
    end
```

```
end

% Patient 5117: Column 4 Zonisamide
ASM(:,1,13)=BIDMC5117(:,4);
% Zonisamide has a max half life of 63 hours, so if there is 5 missed doses
then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,13))
    % Check if the current number is negative
    if ASM(i,1,13) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,13) = 1;
    end
end

% Patient 5146: Column 3 Zonisamide
ASM(:,1,14)=BIDMC5146(:,3);
% Zonisamide has a max half life of 63 hours, so if there is 5 missed doses
then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,14))
    % Check if the current number is negative
    if ASM(i,1,14) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,14) = 1;
    end
end

% Patient 5184: Column 3 Valproic Acid, Column 4 Keppra
ASM(:,1:2,15)=BIDMC5184(:,3:4);
% Depakote has a max half life of 16 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
```

```
for i = 1:length(ASM(:,1,15))
    % Check if the current number is negative
    if ASM(i,1,15) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,15) = 1;
    end
end
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,15))
    % Check if the current number is negative
    if ASM(i,2,15) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,15) = 1;
    end
end

% Patient 5274: Column 3 Lamictal XR, Column 6 Gabapentin, Column 7
% Fycompa
ASM(:,1,16)=BIDMC5274(:,3);
ASM(:,2:3,16)=BIDMC5274(:,6:7);
% Lamictal has a max half life of ~33 hours, so if there are 2 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Loop through each element of the column
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 2;    % Set the limit for consecutive negative numbers
for i = 1:length(ASM(:,1,16))
    % Check if the current number is negative
    if ASM(i,1,16) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,16) = 1;
    end
end
% Gabapentin has a max half life of 7 hours, so if there are 1 missed doses
```

```
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,16))
    % Check if the current number is negative
    if ASM(i,2,16) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,16) = 1;
    end
end
% Fycompa has a max half life of 105 hours, so if there are 8 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 8;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,16))
    % Check if the current number is negative
    if ASM(i,3,16) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,16) = 1;
    end
end


% Patient 5392: Column 3 Vimpat, Column 4 Keppra, Column 5 Depakote, Column
% 6 Zonegran
ASM(:,1:4,17)=BIDMC5392(:,3:6);
% Vimpat has a max half life of ~13 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,17))
    % Check if the current number is negative
    if ASM(i,1,17) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
```

```
    if count >= consecutive_limit
        ASM_HL(i,1,17) = 1;
    end
end
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,17))
    % Check if the current number is negative
    if ASM(i,2,17) < 0
        count = count + 1;   % Increment the count of consecutive negatives
    else
        count = 0;           % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,17) = 1;
    end
end
% Depakote has a max half life of 16 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,17))
    % Check if the current number is negative
    if ASM(i,3,17) < 0
        count = count + 1;   % Increment the count of consecutive negatives
    else
        count = 0;           % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,17) = 1;
    end
end
% Zonisamide has a max half life of 63 hours, so if there is 5 missed doses
then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,4,17))
    % Check if the current number is negative
    if ASM(i,4,17) < 0
        count = count + 1;   % Increment the count of consecutive negatives
    else
        count = 0;           % Reset the count if a positive number is found
    end
```

```
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,4,17) = 1;
    end
end

% Patient 5412: Column 3 Zonegran, Column 4 Trileptal, Column 5 Fycompa
ASM(:,1:3,18)=BIDMC5412(:,3:5);
% Zonisamide has a max half life of 63 hours, so if there is 5 missed doses
then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;              % To keep track of consecutive negative numbers
consecutive_limit = 5;  % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,18))
    % Check if the current number is negative
    if ASM(i,1,18) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,18) = 1;
    end
end
% Oxtellar has a max half life of 2 hours, MHD is 9 hours. So if 1 missed
% dose, then already at risk of being at %50 concentration prior to the next
dose
% Initialize variables
count = 0;              % To keep track of consecutive negative numbers
consecutive_limit = 1;  % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,18))
    % Check if the current number is negative
    if ASM(i,2,18) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,18) = 1;
    end
end
% Fycompa has a max half life of 105 hours, so if there are 8 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;              % To keep track of consecutive negative numbers
consecutive_limit = 8;  % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,18))
    % Check if the current number is negative
```

```
    if ASM(i,3,18) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,18) = 1;
    end
end

% Patient 5507: Column 3 Briviact, Column 4 Lamotrigine, Column 5
% Epidiolex, Column 6 Xcopri
ASM(:,1:4,19)=BIDMC5507(:,3:6);
% Briviact has a max half life of 9 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,19))
    % Check if the current number is negative
    if ASM(i,1,19) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,19) = 1;
    end
end
% Lamictal has a max half life of ~33 hours, so if there are 2 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Loop through each element of the column
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 2;    % Set the limit for consecutive negative numbers
for i = 1:length(ASM(:,2,19))
    % Check if the current number is negative
    if ASM(i,2,19) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,19) = 1;
    end
end
% Epidiolex has a max half life of 61 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
```

```
consecutive_limit = 5;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,19))
    % Check if the current number is negative
    if ASM(i,3,19) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,19) = 1;
    end
end
% XCopri has a max half life of ~60 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;               % To keep track of consecutive negative numbers
consecutive_limit = 5;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,4,19))
    % Check if the current number is negative
    if ASM(i,4,19) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,4,19) = 1;
    end
end

% Patient 5606: Column 3 Vimpat, Column 4 Epidiolex, Column 6 Keppra,
% Column 7 Keppra XR, Column 8 XCopri
ASM(:,1:2,20)=BIDMC5606(:,3:4);
ASM(:,3:5,20)=BIDMC5606(:,6:8);
% Vimpat has a max half life of ~13 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;               % To keep track of consecutive negative numbers
consecutive_limit = 1;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,20))
    % Check if the current number is negative
    if ASM(i,1,20) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,20) = 1;
    end
```

```
end
% Epidiolex has a max half life of 61 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 5;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,20))
    % Check if the current number is negative
    if ASM(i,2,20) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,20) = 1;
    end
end
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,20))
    % Check if the current number is negative
    if ASM(i,3,20) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,20) = 1;
    end
end
% Keppra XR has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,4,20))
    % Check if the current number is negative
    if ASM(i,4,20) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,4,20) = 1;
    end
```

```
end
% XCopri has a max half life of ~60 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 5;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,5,20))
    % Check if the current number is negative
    if ASM(i,5,20) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,5,20) = 1;
    end
end


% Patient 5698: Column 3 Keppra, Column 7 Depakote ER, Column 18 XCopri
ASM(:,1,21)=BIDMC5698(:,3);
ASM(:,2,21)=BIDMC5698(:,7);
ASM(:,3,21)=BIDMC5698(:,18);
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,21))
    % Check if the current number is negative
    if ASM(i,1,21) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,21) = 1;
    end
end
% Depakote has a max half life of 16 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,21))
    % Check if the current number is negative
    if ASM(i,2,21) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
```

```
        count = 0;           % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,21) = 1;
    end
end
% XCopri has a max half life of ~60 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,21))
    % Check if the current number is negative
    if ASM(i,3,21) < 0
        count = count + 1;   % Increment the count of consecutive negatives
    else
        count = 0;           % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,21) = 1;
    end
end

% Patient 5708: Column 9 Keppra, Column 14 XCopri, Column 20 Lamotrigine,
% Column 21 Clobazam, Column 26 Epidiolex
ASM(:,1,22)=BIDMC5708(:,9);
ASM(:,2,22)=BIDMC5708(:,14);
ASM(:,3:4,22)=BIDMC5708(:,20:21);
ASM(:,5,22)=BIDMC5708(:,26);
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,22))
    % Check if the current number is negative
    if ASM(i,1,22) < 0
        count = count + 1;   % Increment the count of consecutive negatives
    else
        count = 0;           % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,22) = 1;
    end
end
% XCopri has a max half life of ~60 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
```

```
consecutive_limit = 5;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,22))
    % Check if the current number is negative
    if ASM(i,2,22) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,22) = 1;
    end
end
% Lamictal has a max half life of ~33 hours, so if there are 2 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Loop through each element of the column
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 2;    % Set the limit for consecutive negative numbers
for i = 1:length(ASM(:,3,22))
    % Check if the current number is negative
    if ASM(i,3,22) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,22) = 1;
    end
end
% Onfi has a max half life of 42 hours, so if there are 3 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 3;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,4,22))
    % Check if the current number is negative
    if ASM(i,4,22) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,4,22) = 1;
    end
end
% Epidiolex has a max half life of 61 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
```

```
consecutive_limit = 5;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,5,22))
    % Check if the current number is negative
    if ASM(i,5,22) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,5,22) = 1;
    end
end

% Patient 5731: Column 3 Carbamazepine, Column 4 Levetiracetam, Column 5
% Vimpat, Column 6 repeat Vimpat?? will ignore for now given no negative
% values/missed doses anyways
ASM(:,1:3,23)=BIDMC5731(:,3:5);
% Carbamazepine has a max half life of 65 hours, so if there are 5 missed
doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;               % To keep track of consecutive negative numbers
consecutive_limit = 5;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,23))
    % Check if the current number is negative
    if ASM(i,1,23) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,23) = 1;
    end
end
% Keppra has a max half life of 8 hours, so if there is 1 missed dose then
% already at risk of being at %50 concentration prior to the next dose
% Initialize variables
count = 0;               % To keep track of consecutive negative numbers
consecutive_limit = 1;   % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,23))
    % Check if the current number is negative
    if ASM(i,2,23) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,23) = 1;
```

```
    end
end
% Vimpat has a max half life of ~13 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,23))
    % Check if the current number is negative
    if ASM(i,3,23) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,23) = 1;
    end
end

% Patient 5782: Column 3 Vimpat, Column 6 XCopri
ASM(:,1,24)=BIDMC5782(:,3);
ASM(:,2,24)=BIDMC5782(:,6);
% Vimpat has a max half life of ~13 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,24))
    % Check if the current number is negative
    if ASM(i,1,24) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,24) = 1;
    end
end
% XCopri has a max half life of ~60 hours, so if there are 5 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                 % To keep track of consecutive negative numbers
consecutive_limit = 5;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,24))
    % Check if the current number is negative
    if ASM(i,2,24) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
```

```
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,24) = 1;
    end
end

% Patient 5790: Column 3 Keppra
ASM(:,1,25)=BIDMC5790(:,3);
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,25))
    % Check if the current number is negative
    if ASM(i,1,25) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,25) = 1;
    end
end

% Patient 5799: Column 3 Keppra, Column 5 Tegretol XR
ASM(:,1,26)=BIDMC5799(:,3);
ASM(:,2,26)=BIDMC5799(:,5);
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 1;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,26))
    % Check if the current number is negative
    if ASM(i,1,26) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,26) = 1;
    end
end
% Carbamazepine has a max half life of 65 hours, so if there are 5 missed
doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                % To keep track of consecutive negative numbers
consecutive_limit = 5;    % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,26))
    % Check if the current number is negative
    if ASM(i,2,26) < 0
        count = count + 1;  % Increment the count of consecutive negatives
```

```
    else
        count = 0;           % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,26) = 1;
    end
end

% Patient 59589: Column 3 Topamax, Column 4 Briviact, Column 5 Vimpat
ASM(:,1:3,27)=BIDMC59589(:,3:5);
% Topomax has a max half life of 23 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,1,27))
    % Check if the current number is negative
    if ASM(i,1,27) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;           % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,1,27) = 1;
    end
end
% Briviact has a max half life of 9 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,2,27))
    % Check if the current number is negative
    if ASM(i,2,27) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;           % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,2,27) = 1;
    end
end
% Vimpat has a max half life of ~13 hours, so if there are 1 missed doses
% consecutively, then at risk of being at ~50% concentration.
% Initialize variables
count = 0;                  % To keep track of consecutive negative numbers
consecutive_limit = 1;     % Set the limit for consecutive negative numbers
% Loop through each element of the column
for i = 1:length(ASM(:,3,27))
```

```
    % Check if the current number is negative
    if ASM(i,3,27) < 0
        count = count + 1;  % Increment the count of consecutive negatives
    else
        count = 0;          % Reset the count if a positive number is found
    end
    % If we have reached the limit of consecutive negatives, mark the row
    if count >= consecutive_limit
        ASM_HL(i,3,27) = 1;
    end
end
```

Matrix of missed doses

The raw data for the ASM data is in half days. Let's similarly shrink the temporal resolution such that we look at seizure risk from missed ASMs as days rather than half days.

```
[numRows, numCols, numSubs] = size(ASM_HL);
Missed_Doses = zeros(numRows / 2, numCols, numSubs);
    for row = 1:numRows/2
        for col = 1:numCols
            for subs = 1:numSubs
            if ASM_HL(2*row-1, col, subs) == 1 || ASM_HL(2*row, col, subs)
== 1
                  Missed_Doses(row, col, subs) = 1;
            else
                   Missed_Doses(row,col,subs)= 0;
            end
        end
        end
    end

for j = 1:27
     Storage=Missed_Doses(:,:,j);
     Missed_Doses_Periods(:,j)=max(Storage,[],2);
end

% Let's create a way to determine how many days back we look at for missed
% doses.
Window_MissedDoses = 0; %set window length here in days. 0 is the starting
value (day prior)
for i = 90:280
    for j = 1:27
        k = i-Window_MissedDoses;
    Missed_Doses_Periods2(i,j)=sum(Missed_Doses_Periods(k:i,j));
    end
end
Missed_Doses_Periods_clean=Missed_Doses_Periods2(90:280,:);
```

Organize the data for mixed effects analysis

Let's organize the data first into a table and concatenate the variables vertically. 191 estimates x 27 subjects = 5157.

```
MAConcat=reshape(MA_90,[5157,1]); %concatenate the MA data into a vector
MissedDosesConcat=reshape(Missed_Doses_Periods_clean,[5157,1]); %concatenate
the missed ASMs into a vector
SeizureDaysConcat=reshape(seizure_days_forecast,[5157,1]); %concatenate the
seizure occurence data
%Let's create a subject ID that corresponds to each of the entries above
for i = 1:27;
    SubjectID(1:191,i)=i;
end
SubjectIDConcat=reshape(SubjectID,[5157,1]);
```

Feature Scaling with Fisher Transform

We apply it to the MA data which is a proportion in the range [0, 1]. Note: The MissedDoses variable is a binary categorical variable, so no scaling is needed for it.

```
FisherTransformedMA = atanh(MAConcat);
```

Place the data into a single table

The table now includes the transformed MA data. Table will essentially have one column consisting of MA, missed doses the day prior, seizures on day window+1, and subject ID

```
tbl = table(MissedDosesConcat, FisherTransformedMA, SubjectIDConcat,
SeizureDaysConcat, 'VariableNames', {'MissedDoses_tbl', 'MA_tbl', 'ID_tbl',
'Seizure_tbl'});
```

Fit the mixed effects binomial model:

```
%Seizure on day window+1 is the response variable (0 = no seizure, 1 =
%seizure)
%Fisher transformed MA is a fixed effect/predictor variable
%MissedDoses is a fixed effect/predictor variable (0 = above one half life,
1 = below one half life)
%Subject ID is the grouping variable/random effect (range 1-27)
lme=fitglme(tbl,['Seizure_tbl ~ MissedDoses_tbl + MA_tbl + (1|
ID_tbl)'],'Distribution','Binomial');
```

Display model summary

This command displays a detailed summary of the fitted mixed effects model, including coefficient estimates, p-values, and model fit statistics.

```
disp(lme);
```

Compare full model to null model using deviance

As requested, this section manually calculates the deviance difference and transforms it to a chi-squared test statistic. A small p-value indicates that the full model is a significantly better fit than the null model.

```
% First, fit a simpler "null" model that only includes the intercept
% and the random effect.
lme_null = fitglme(tbl, ['Seizure_tbl ~ 1 + (1|ID_tbl)'], 'Distribution',
'Binomial');

% Calculate the difference in deviance between the two models using the log-
likelihood.
% The deviance is -2 * log-likelihood, so the difference is -2 *
(logLikelihood_full - logLikelihood_null).
% This difference follows a chi-squared distribution.
deviance_diff = lme_null.LogLikelihood - lme.LogLikelihood;
deviance_diff = -2 * deviance_diff;

% The degrees of freedom for the test is the difference in the number
% of fixed-effects parameters (3 for the full model, 1 for the null model).
df = 2;

% Calculate the p-value using the chi-squared cumulative distribution
function.
p = 1 - chi2cdf(deviance_diff, df);

% Display the deviance difference and p-value
fprintf('The deviance difference between the full model and the null model
is: %.4f\n', deviance_diff);
fprintf('The p-value for the chi-squared test is: %.4f\n', p);
```

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